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Controlling the transmission of broadband light through scattering media using a digital micromirror device

RABISANKAR SAMANTA AND SUSHIL MUJUMDAR* 

Nano-Optics and Mesoscopic Optics Laboratory, Tata Institute of Fundamental Research, 1, Homi Bhabha Road, Mumbai, 400 005, India
*mujumdar@tifr.res.in

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Wavefront shaping has emerged as a valuable technique in complex photonics, wherein the various eigenmodes of the disordered medium are selectively excited to control the overall transmission through the medium. The process utilizes active optical devices such as liquid crystal-based spatial light modulators (LC-SLM), deformable mirrors (DM), and digital micromirror devices (DMD). Among these, the latter is preferred for imaging through dynamic scattering media such as living biological tissues due to their high-speed refresh rate and increased resolution. This study employs a genetic algorithm along with binary amplitude modulation generated by a digital micromirror device to spatially and spectrally control the large spectral bandwidth through a scattering medium. We illustrate spatial single-point focusing of broadband light, multipoint focusing of broadband light, and programmable spectral filtering of the same through disordered samples. © 2023 Optica Publishing Group

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Distortion in light waves induced by randomly varying environments is a common occurrence in nature and is actively addressed by external corrections. In astronomy, turbulence in the atmosphere lowers the image quality of any astronomical object [1]. In medical physics, biological structures play the same role of distorting light. For example, the activity of neurons embedded inside living tissue becomes intractable to noninvasive techniques [2]. In ophthalmology, the imperfect optics in the eyes limit retinal imaging at a requisite resolution [2]. For several decades, astronomers have been using the adaptive optics (AO) approach to correct the wavefront distortion [1] caused by turbulence. This approach has also been adopted in medical science problems [2] and proved to be successful. However, adaptive optics has limitations in terms of the strength of distortions. In terms of disorder, the scattering mean free path of light in the above-mentioned examples remains rather large, resulting only in distortion of the image, but not its elimination. In comparison, in strongly disordered structures, the mean free path of light is small enough to completely bury an image in a sea of speckles. In such cases, wavefront shaping techniques have to be employed which utilize the information of the actual

configuration of disorder. By shaping the phase of the input wavefront with the help of a liquid crystal-based spatial light modulator (LC-SLM), it was possible to focus light through a strongly scattering medium [3]. This technique allowed many possibilities of wavefront shaping in scattering media leading to the creation of a perfect lens [4] or a pulse-shaper [5,6]. Toward focusing through a scattering medium, various approaches such as a sequential algorithm [3,7], optical phase conjugation [8], genetic algorithm [9], and transmission matrix [10] were utilized. In the case of a spatial light modulator (SLM), all channels in the scattering sample are excited with altered phases. Constructive interference at the point of interest in image space is enhanced, while concomitantly, destructive interference in the rest of the image is facilitated. Apart from spatial modifications, spectral (and hence temporal), modifications have been demonstrated. An ultrashort pulse with a broadband spectrum creates spatiotemporal and even spatio-spectral speckle patterns in a random medium. An SLM was used to minimize spatial and temporal distortions [5,6]. Similarly, spatiotemporal focusing of light was achieved through a random medium [11], and in another application, the spectrally resolved point-spread-function thereof was engineered [12]. On the other hand, the multi-spectral transmission matrix of a scattering medium was estimated for spatiotemporal controlling of broadband light [13,14]. A combination of an SLM and a random medium can act as a programmable spectral filter [15–18]. Engineering the pump beam, researchers were able to control the spectral emission of random lasers [19–21], micro-disk lasers [22], and even broadband second-harmonic light generation in nonlinear random media [23].

Although LC-SLMs are versatile elements offering a variety of applications and have the ability to easily modulate the phase of a wavefront, they operate at low speeds and suffer from resolution challenges, both conditions arising from the natural molecular properties of liquid crystals. On the other hand, a digital micromirror device (DMD) is an entirely mechanical device comprising of an array of microns-sized mirrors, wherein each mirror is accessible for independent control. The utility of a DMD in ophthalmology is already being investigated [24]–[26]. Such DMDs can be as fast as ~ 30 kHz and are suitable for high-speed applications such as focusing through a dynamic scattering medium [27]. DMDs are maximally used

for binary amplitude modulation. Binary amplitude modulation is essentially the process of switching ON and OFF selected transmission channels of the scattering matrix of the medium. This essentially controls the interference of the relevant channel outputs at the target locations. Effectively, the optical power admitted into the sample is reduced. DMD can be used as a phase modulator by using the Lee holography method [27], although the process is lossy. The ability to control broadband light with high-speed modulation will be an impactful tool in various applications. However, performing experiments with a high-intense pulsed laser can damage the DMD pixels [28]. In this work, we demonstrate active, on-demand control of modes of an ultrashort pulsed laser (large spectral bandwidth) through a scattering medium using a DMD as a binary amplitude modulator. A genetic algorithm (GA) was employed for achieving the wavefront correction using the ON–OFF switching of transmission channels. The input wavefront was optimized to get a single and double focus after the scattering medium. Further, we have exploited the DMD and scattering medium as a programmable spectral filter. In other words, we can enhance any arbitrary wavelength, or even multiple wavelengths at any point in the speckle field.

Toward the experiment [see Fig. 1(a)], we employ an ultrashort pulse beam of a Ti:Sapphire laser (repetition rate 80 MHz, specified pulse width 75 fs, central wavelength ~ 812 nm [see Fig. 1(a) inset] whose full width half maximum (FWHM) is approximately 10 nm, with the total spectrum ranging from ~ 803 to ~ 823 nm as our broadband source. Given the sensitivity of a speckle pattern to even subnanometer deviations in wavelength, this beam qualifies as broadband in these experiments [29,30]. The beam is expanded to fill the entire chip area of the DMD (Texas Instruments, Vialux module V650L NIR), which has a pitch size of $10.8\ \mu\text{m}$ and a resolution of 1280×800 pixels. The exact orientation of the DMD with respect to the optical setup is described in Supplement 1. The light reflected from the DMD is collimated using a combination of two lenses ($L1 = 30$ cm, $L2 = 20$ cm, plano-convex) and then focused onto the scattering medium with a $10\times$ magnification objective (NA 0.28). We used an aperture to select only the zeroth-order diffraction from the DMD. The scattering medium is a light shaping diffuser (Newport, 10DKIT-C3) with a diffusion angle (FWHM) of 80° . Finally, the scattered light is collected by another objective with a $50\times$ magnification (NA 0.55), and directed toward a CCD camera (DCU223M, Thorlabs) with the aid of a tube lens (focal length 20 cm). Each pixel of the CCD has an area of $4.65 \times 4.65\ \mu\text{m}^2$. The DMD employed in our experiment consists of 1280×800 pixels. Among these, we utilize 780×780 pixels picked in the central region of the DMD. We group 10×10 pixels into one segment which eventually provides 78×78 which is 6084 active segments. We control the DMD using a well-known PYTHON module ALP4lib [31]. A GA [32] was employed to modulate the input light into the scattering medium [9]. A similar GA has been employed earlier for binary wavefront optimization [33,34].

Genetic algorithm. At the start, we generate a series of $N = 200$ parent masks of size 78×78 containing random binary values of 0 or 1. Each mask is loaded serially onto the DMD and the transmitted intensity distribution is simultaneously captured as images by the CCD. The intensity value at the target pixel (i, j) on the CCD determines the fitness of the mask, subsequent to which all N masks are sorted with decreasing fitness. As a result, parents creating higher intensity values $I_{(i,j)}$ are accorded

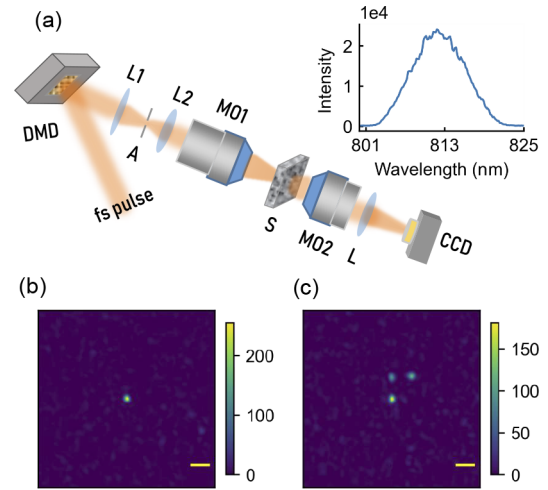


Fig. 1. (a) Schematic of the experimental setup for broadband light focusing and spectral filtering through a scattering medium: DMD, digital micromirror device; MO1, $10\times$ microscope objective [numerical aperture (NA) 0.28]; MO2, $50\times$ microscope objective (NA 0.55); S, scattering medium; ($L1$, $L2$, L), lens; A, aperture; CCD, charged couple device, which is replaced with an optical fiber for spectral filtering. Inset is the spectral band of the femtosecond laser. (b) Conventional focusing through the scattering medium (light shaping diffuser) with monochromatic light was used to benchmark the algorithm, for single-point focus (Visualization 1) and (c) multi-point focus (Visualization 2). Scale bar: $3.7\ \mu\text{m}$.

higher ranks. Next, the breeding process is initiated where we generate a number G (typically $N/2$) of offspring from the parent masks. The breeding process involves a natural selection algorithm wherein two parents $F_{78 \times 78}$ and $M_{78 \times 78}$ are selected preferentially with higher rankings. A random binary template $T_{78 \times 78}$ is generated that combines with the parents and generates an offspring mask given by $O = F \cdot T + M \cdot (1 - T)$, where the $(\cdot)$ indicates elementwise multiplication. A fraction of elements of O is then mutated, as in, some of the elements are randomly switched from 1 to 0 or *vice versa*. In this way, we generate $G = N/2$ numbers of offspring. These new offspring masks G are loaded onto the DMD and fitness values based on $I_{(i,j)}$ are obtained. These G masks are used to swap the lowest-ranked $N/2$ masks in the parent series, which is then re-sorted based on the fitness values. This completes one iteration creating a new parent series. After several iterations, the algorithm generates a suitable mask that gives a single-point/multi-point focus on the camera. The parameters used for the GA are (a) initial mutation rate: 0.01; (b) final mutation rate, 0.0025; and (c) decay factor in the mutation, 650 [33]. We define the enhancement (η) as the ratio of intensity at the target pixel after optimization to the average intensity for uniform illumination. Essentially, $\eta = I_{ij}^a / \langle I^{uni} \rangle$, where I_{ij}^a is the intensity at the target pixel [let us say $(i, j)^{\text{th}}$ pixel] after each iteration. Toward bench-marking of the algorithm, we conducted a focusing experiment employing monochromatic He-Ne laser light through our scattering medium. The enhancement of intensity at a single point and multiple points is illustrated in Figs. 1(b) and 1(c), respectively.

In Fig. 2 we illustrate the focusing of the broadband light through the diffuser. Figure 2(a) is the speckle for uniform illumination, while Fig. 2(c) is the optimized image after 180 iterations. A clear bright spot is seen to appear at our target

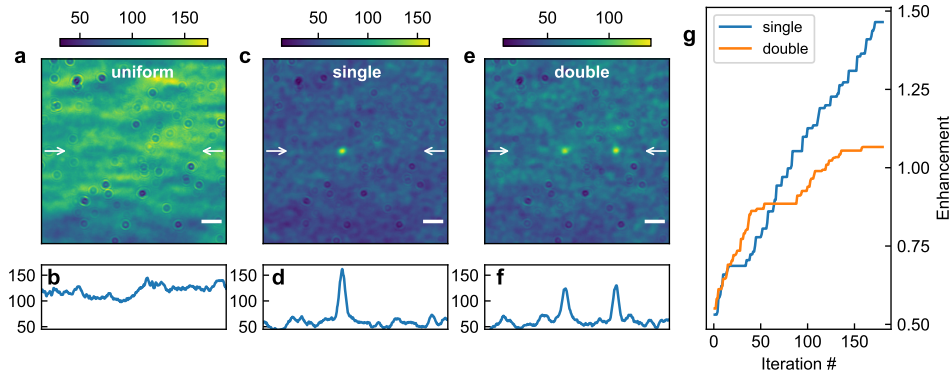


Fig. 2. Focusing of broadband light through the scattering medium. (a) Speckle pattern on the CCD after uniform light illumination from the DMD onto the sample. (c) Top-ranked pattern after 180 iterations, showing optimized focusing at one point (Visualization 3). (e) Same, at two points (Visualization 4). Scale bar: 3.7 μm . Panels (b), (d), and (f) are the intensity plots measured in the middle row of pixels (white arrows) in (a), (c), and (e), respectively. (g) Enhancement factor with the iteration number.

location, accompanied by a reduction in speckle contrast in the remaining image. This process is inhibited by the broadband nature of light that limits the speckle contrast, which explains the lower enhancement compared with monochromatic light. Next, we proceeded to multipoint focusing on the same sample. The feedback function chosen for the GA is $f(I_1, I_2) = \frac{(I_1 + I_2)}{2} [\text{sinc}(\frac{I_1}{I_2} - 1)]^2$, where I_1 and I_2 are the intensities at the two points, respectively. This function ensures that the intensity at both points gains concomitantly. This avoids a situation where one peak cannibalizes the other, such as in cases where the feedback parameter is the mean of the two intensities, as exemplified in Supplement 1 (Figs. S1 and S2, Visualization 5). Toward multipoint optimization, a generic form of the function can be written as

$$f(I_1, \dots, I_N) = \bar{I} \prod_{i=1}^{N-1} \text{sinc}^2 \left[\frac{I_i}{I_{i+1}} - 1 \right], \quad (1)$$

where $\bar{I} = (\sum_{i=1}^N I_i)/N$, and I_i are the intensities at the points x_i on the CCD. Figure 2(e) shows the result for two-point optimization after 180 iterations wherein the signatures of two-point enhancement are visible. Evidently, multipoint focusing is a highly constrained problem with broadband light. Figures 2(b), 2(d), and 2(f) at the bottom are the intensity plots measured in the middle row of pixels (indicated by the white arrows) of Figs. 2(a), 2(c), and 2(e), respectively. Figure 2(g) depicts the enhancement η with iteration number where the blue and orange lines are for single and multi-point focusing, respectively. Next, we implemented the spectral filtering experiments using binary amplitude modulation by the DMD. A multi-mode fiber (core ϕ 50 μm , NA 0.22) is positioned in the speckle field, with the other end connected to a spectrometer (OceanOptics USB4000, resolution 0.19 nm). The spectral intensity of the target wavelength is used as a feedback parameter for the GA, whose operating parameters were maintained the same. The final result is demonstrated in Fig. 3, where the red line represents the initial unoptimized spectrum, while the blue line shows the spectrum under optimized illumination. The optimized spectral peak at the target location (gray dashed vertical line) has an FWHM of ~ 0.81 nm, essentially the resolution of our setup. The peaks at other wavelengths are almost totally suppressed. The intensity enhancement is plotted in the inset, which reaches saturation after approximately 40 iterations. An enhancement of approximately 2.35 is observed at the target wavelength for

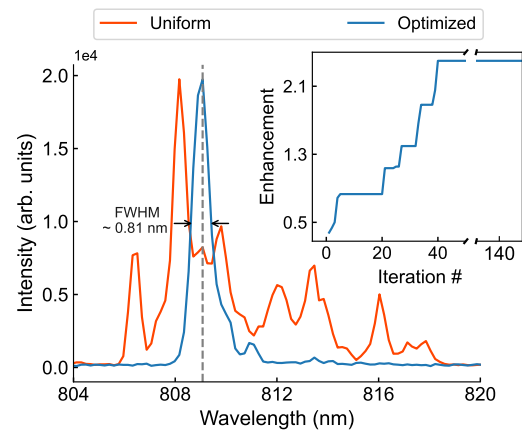


Fig. 3. Single wavelength optimization (Visualization 6). The red line describes the spectral speckle pattern for uniform illumination. The blue line is the optimized spectrum that peaked at 809.09 nm after 150 iterations with an FWHM of ~ 0.81 nm. Here, the dashed gray line indicates the position of the target wavelength. Inset is the enhancement of the intensity at the target wavelength at each iteration.

this run with a rejection ratio of 6.45 dB (which represents reduced transmission in decibels at the rejected wavelengths). This enhancement number is not absolute and can differ for different wavelengths. The very discrete ladder (inset) with large jumps shows that several generations have to pass before a consequential mutation takes place that is impactful and realizes a substantial enhancement. The ability of the DMD to selectively choose channels that can filter light spectrally is a promising feature toward high-speed spectral filtering.

Bimodal spectral filtering is investigated by selecting two wavelengths (say λ_1 and λ_2) in the initial spectrum for maximization. The feedback function chosen for the GA is the same as in spatial focusing, except here, I_1 and I_2 are the intensities at λ_1 and λ_2 , respectively. This function is designed to maximize both wavelengths with equal weights. (Supplement 1, Figure S3 and Visualization 7 show the cannibalization effect when the mean intensity is used for feedback.) Figure 4 presents our findings for dual-wavelength optimization, where the spectrum after 180 iterations shows two distinct peaks at the target wavelengths with resolution-limited bandwidth. The mean intensity between

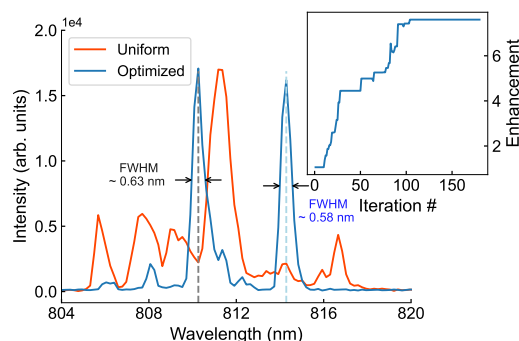


Fig. 4. Double wavelength optimization (Visualization 8). The red line describes the spectral speckle pattern for uniform illumination. Two vertical dashed lines positioned at 810.26 nm (gray color, FWHM ~ 0.63 nm) and 814.29 nm (light blue color, FWHM ~ 0.58 nm) are the target wavelengths. The blue line is the optimized spectrum after 180 iterations. Inset is the mean enhancement at the target wavelengths after each iteration.

the peaks is enhanced by a factor of 7.56 compared with the mean intensity at the same wavelengths in the unoptimized spectrum while the rejection ratio at the rejected wavelengths comes out to be 5.03 dB. Careful inspection of the enhancement process shows that the algorithm picks up generations that alternatively enhance one wavelength after another. See Supplement 1, Fig. S4. These data endorse that the scattering medium and the DMD can operate as an efficient programmable spectral filter [15,17].

In conclusion, we have demonstrated spatio-spectral control of broadband light through a scattering medium using a DMD, which employs a binary amplitude modulation process wherein select transmission channels are switched ON, and others are OFF. A genetic algorithm is applied for the optimization of the process. Spatial focusing on a single point or two points in the output plane is achieved after optimizing the transmission channels. Next, a combination of DMD and a scattering medium is exploited as a programmable spectral filter, wherein the transmission of a single sharp-peaked spectrum as well as a dual-peaked spectrum is demonstrated. The width of the peaks was in the range between 0.55 nm to 0.85 nm, limited by the resolution of the spectrometer. These experiments provide convincing evidence of the applicability of a DMD in wavefront modulation for broadband light. It can be envisaged that the DMD will allow high-speed optimisation in the presence of appropriate accessory electronics. For instance, the CCD was employed for visualizing the focusing process. A simple replacement of the CCD by an APD at the desired site reduced the time of optimization by a factor of 5. See Supplement 1, Figs. S5 and S6. Earlier, a MEMS-based wavefront shaper (about 4 kHz refresh rate) was used in conjunction with FPGA and fast electronics to achieve focusing through dynamic scattering media [35]. A DMD (about 30 kHz refresh rate) would be much faster in the same scenario. Overall, DMD-based spectral filtering will turn out to be vital for many applications like hyperspectral imaging, selective pumping inside biological tissue, transmission through dynamic disorder, spectral filtering in turbulent media, etc.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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